Mac - Install Development Tools

In this guide, we will install the following development tools

- Visual Studio Code
- nvm
- node
- npm
- tsc

Versions

This course has been tested with the following software versions:

- nvm 0.35.0
- npm 8.15
- node 16.17.0
- tsc 4.7.4
- React 18.2.0
- Spring Boot 2.7.3

It is **highly recommended** that you use the versions listed above to make sure you do not encounter any issues with the course. If you choose to use other versions then the code may not work as expected.

Prerequisites

- git: You must have the git command-line tool installed.
    - To install git, visit: https://git-scm.com/

Install Visual Studio Code

Visual Studio Code is a general-purpose IDE that support many programming languages. Visual Studio code has built-in support for TypeScript.

1. In a web browser, visit https://code.visualstudio.com/
2. Follow the link to download Visual Studio Code for Mac
3. Unzip the downloaded file
4. Move the Visual Studio Code app to your Applications Folder
5. In your Applications Folder, start the Visual Studio Code app

Install NVM

nvm is the Node Version Manager. It allows you to install multiple versions of Node on your computer. Once installed, you can easily switch to a different version of Node. This is very useful if you want to develop/test using different versions of Node. When you are developing multiple projects, you will most likely need to support different versions of Node.

The best feature of nvm is that it makes the installation of Node very easy for Mac and Linux systems. Historically, to install Node on Mac/Linux, you would have to make frequent use of the "sudo" command to elevate privileges. This was a painful and clunky process during development. Thanks to nvm, we no longer have to use "sudo" when using Node.

Also, the Node ecosystem is a fast-moving target, and versions change frequently. nvm makes it easy to stay up to date with the latest version of Node.

The website for nvm is: https://github.com/nvm-sh/nvm

Note: You must have git installed before running this script.

Install

1. Open a new terminal window
2. Enter the following commands:

```
curl -o- https://raw.githubusercontent.com/creationix/nvm/v0.35.0/install.sh | bash
```

The script clones the nvm repository to `~/.nvm` and adds the source line to your profile: `~/.bashrc`.

You will see output similar to the following:

```
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 13527  100 13527    0     0  62065      0 --:--:-- --:--:-- --:--:-- 62336
=> Downloading nvm from git to '/Users/luv2code/.nvm'
=> Cloning into '/Users/luv2code/.nvm'...
remote: Enumerating objects: 286, done.
...

=> Appending nvm source string to /Users/luv2code/.bashrc
=> Appending bash_completion source string to /Users/luv2code/.bashrc
=> Close and reopen your terminal to start using nvm or run the following to use it
now:

...
```

Setting Up Paths

Only for Z Shell (zsh) users

3. If you are using Z shell (zsh), apply the new PATH settings with:

   ```
   code ~/.zshrc
   ```

- The `code` command will launch Visual Studio Code.
- Add the following content to the `.zshrc` file

```
export NVM_DIR="$HOME/.nvm"
[ -s "$NVM_DIR/nvm.sh" ] && \. "$NVM_DIR/nvm.sh"  # This loads nvm
[ -s "$NVM_DIR/bash_completion" ] && \. "$NVM_DIR/bash_completion"  # This loads nvm bash_completion
```

- Save the file and exit the text editor.
- Apply the changes with the following command

```
source ~/.zshrc
```

Only for BASH shell users

3. If you are using BASH shell, apply the new PATH settings with:

   ```
   source ~/.bashrc
   ```

4. Update your `~/.bash_profile` file to reference your `~/.bashrc` file.

   ```
   code ~/.bash_profile
   ```

   The `code` command will launch Visual Studio Code.

5. Your `.bash_profile` file should now be open in Visual Studio Code.
6. Add the following text to your `.bash_profile` file:

```
if [ -r ~/.bashrc ]; then
   source ~/.bashrc
fi
```

7. Save the file and exit Visual Studio Code.

Verify the Installation

8. Move back to your terminal window.
9. Verify the installation by typing the following command:

   ```
   nvm --version
   ```

   If the installation is successful, you will see the version number.

For details on other nvm commands, use `nvm --help` or see the docs: https://github.com/nvm-sh/nvm

Install Node

Node is the runtime environment for executing JavaScript code from the command-line. By using Node, you can create any type of application using JavaScript including server-side / backend applications.

In this course, we'll use Node to run applications that we develop using TypeScript and Angular.

1. In your terminal window, type the following command:

   ```
   nvm install node
   ```

   You will see the following output

   ```
   Downloading and installing node v18.4.0...
   Downloading https://nodejs.org/dist/v18.4.0/node-v18.4.0-darwin-x64.tar.xz...
   ######################################################################## 100.0%
   Computing checksum with shasum -a 256
   Checksums matched!
   Now using node v18.4.0 (npm v8.12.1)
   Creating default alias: default -> node (-> v18.4.0)
   ```

2. Verify the Node installation

```
node --version
```

If the installation is successful, you will see the version number

Note: The Node installation also includes npm (Node Package Manager).

3. Verify npm is installed

   ```
   npm --version
   ```

   If the installation is successful, you will see the version number.

   Note: Node will have a different number than npm. This is similar to a different Java JDK version number compared to the Maven version number.

   In this example, Node is similar to the Java JDK. And npm is similar to Maven.

Switch Node Versions

Note: This course has been tested with Node 16.17. We will install this version.

1. Install and use Node 16.17

```
nvm install 16.17.0
nvm use 16.17.0
```

Install tsc

tsc is the TypeScript compiler. We use tsc to compile TypeScript code into JavaScript code. We can install the TypeScript compiler using the Node Package Manager (npm)

Note: This course has been tested with TypeScript 4.7. We will install this version.

1. In your terminal window, enter the following command

```
npm install --location=global typescript@4.7.4
```

The `--location=global` installs this as a global package. The TypeScript compiler will be available to all directories for this user.

You can verify the installation

```
tsc --version
```

If the installation is successful, you will see the version number.

That's it! You have successfully installed the development tools: Visual Studio Code, nvm, node, npm and tsc.